

QFT Dimensional Analysis

In $\hbar = c = 1$ units, all quantities are measured in units of energy to some power. For example $[m] = [p^\mu] = E^{+1}$ while $[x^\mu] = E^{-1}$ where $[m]$ stands for the *dimensionality* of the mass rather than the mass itself, and ditto for the $[p^\mu]$, $[x^\mu]$, *etc.* The action

$$S = \int d^4x \mathcal{L}$$

is dimensionless (in $\hbar \neq 1$ units, $[S] = \hbar$), so the Lagrangian of a 4D field theory has dimensionality $[\mathcal{L}] = E^{+4}$.

Canonical dimensions of quantum fields follow from the free-field Lagrangians. A scalar field $\Phi(x)$ has

$$\mathcal{L}_{\text{free}} = \frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi - \frac{1}{2} m^2 \Phi^2, \quad (1)$$

so $[\mathcal{L}] = E^{+4}$, $[m^2] = E^{+2}$, and $[\partial_\mu] = E^{+1}$ imply $[\Phi] = E^{+1}$. Likewise, the EM field has

$$\mathcal{L}_{\text{free}}^{\text{EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \implies [F_{\mu\nu}] = E^{+2}, \quad (2)$$

and since $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, the $A_\nu(x)$ field has dimension

$$[A_\nu] = [F_{\mu\nu}] / [\partial_\mu] = E^{+1}. \quad (3)$$

The massive vector fields also have $[A_\nu] = E^{+1}$ so that both terms in

$$\mathcal{L}_{\text{free}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m^2 A_\nu A^\nu \quad (4)$$

have dimensions $[F^2] = [m^2 A^2] = E^{+4}$.

In fact, all *bosonic* fields in 4D spacetime have canonical dimensions E^{+1} because their kinetic terms are quadratic in $\partial_\mu(\text{field})$. On the other hand, the fermionic fields like the Dirac field $\Psi(x)$ with free Lagrangian

$$\mathcal{L}_{\text{free}} = \bar{\Psi}(i\gamma^\mu \partial_\mu - m)\Psi \quad (5)$$

have kinetic terms with two fields but only one ∂_μ . Consequently, $[\mathcal{L}] = E^{+4}$ implies $[\bar{\Psi}\Psi] = E^{+3}$ and hence $[\Psi] = [\bar{\Psi}] = E^{+3/2}$. Similarly, all other types of fermionic fields in 4D have canonical dimension $E^{+3/2}$.

In QFTs in other spacetime dimensions $d \neq 4$, the bosonic fields such as scalars and vectors have canonical dimension

$$[\Phi] = [A_\nu] = E^{+(d-2)/2} \quad (6)$$

while the fermionic fields have canonical dimension

$$[\Psi] = E^{+(d-1)/2}. \quad (7)$$

In perturbation theory, dimensionality of coupling parameters such as λ in $\lambda\Phi^4$ theory or e in QED follows from the field's canonical dimensions. For example, in a 4D scalar theory with Lagrangian

$$\mathcal{L} = \frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi - \frac{1}{2}m^2\Phi^2 - \sum_{n \geq 3} \frac{C_n}{n!}\Phi^n, \quad (8)$$

the coupling C_n of the Φ^n term has dimensionality

$$[C_n] = [\mathcal{L}] / [\Phi]^n = E^{4-n}. \quad (9)$$

In particular, the cubic coupling C_3 has positive energy dimension E^{+1} , the quartic coupling $\lambda = C_4$ is dimensionless, while all the higher-power couplings have negative energy dimensions E^{negative} .

Now consider a theory with a single coupling g of dimensionality $[g] = E^\Delta$. The perturbation theory in g amounts to a power series expansion

$$\mathcal{M}(\text{momenta}, g) = \sum_N \left(\frac{g}{\mathcal{E}^\Delta} \right)^N \times F_N(\text{momenta}) \quad (10)$$

where $\mathcal{E}$ is the overall energy scale of the process in question and all the F_N functions of momenta have the same dimensionality. The power series (10) is asymptotic rather than convergent, so it makes sense only when the expansion parameter is small,

$$\frac{g}{\mathcal{E}^\Delta} \ll 1. \quad (11)$$

For a dimensionless coupling g , this condition is simply $g \ll 1$, but for $\Delta \neq 0$, the situation is more complicated.

For couplings of positive dimensionality $\Delta > 0$, the expansion parameter (11) is always small for high-energy processes with $\mathcal{E} \gg g^{1/\Delta}$. But for low energies $\mathcal{E} \lesssim g^{1/\Delta}$, the expansion parameter becomes large and the perturbation theory breaks down. This is a major problem for theories with $\Delta > 0$ couplings of *massless particles*. However, if all the particles participating in a $\Delta > 0$ coupling are massive, then all processes have energies $\mathcal{E} \gtrsim M_{\text{lightest}}$, and this makes couplings with $\Delta > 0$ OK as long as

$$g \ll M_{\text{lightest}}^\Delta. \quad (12)$$

Couplings of negative dimensionality $\Delta < 0$ have an opposite problem: The expansion parameter (11) is small at low energies but becomes large at high energies $\mathcal{E} \gtrsim g^{-1/\Delta}$. Beyond the maximal energy

$$E^{\text{max}} \sim g^{1/(-\Delta)}, \quad (13)$$

the perturbation theory breaks down and we may no longer compute the S-matrix elements $\mathcal{M}$ using any finite number of Feynman diagrams.

Worse, in Feynman diagrams with loops one must worry not only about momenta k^μ of the incoming and outgoing particles but also about momenta q^μ of the internal lines. Basically, an L -loop diagram contributing to N^{th} term in the expansion (10) produces something like

$$g^N \times \int d^{4L} q \mathcal{F}_N(q, k, m) \quad \text{where} \quad [\mathcal{F}_N] = E^{-N\Delta-4L+C}, \quad C = \text{const.} \quad (14)$$

For very large loop momenta $q \gg k, m$, dimensionality implies $\mathcal{F}_N \propto q^{-N\Delta-4L+C}$, so for $N(-\Delta) + C \geq 0$, the integral (14) diverges as $q \rightarrow \infty$. Moreover, the degree of divergence increases with the order N of the perturbation theory, so any scattering amplitude becomes divergent at high orders. Therefore, *field theories with $\Delta < 0$ couplings do not work as complete theories*.

However, theories with $\Delta < 0$ may be used as approximate *effective theories* (without the divergent loop graphs) for low-energy processes, $\mathcal{E} \lesssim \Lambda$ for some $\Lambda < g^{-1/\Delta}$. For example,

the Fermi theory of weak interactions

$$\mathcal{L}_{\text{int}} = \frac{G_F}{\sqrt{2}} \sum_{\substack{\text{appropriate} \\ \text{fermions}}} \bar{\Psi} \gamma_\mu (1 - \gamma^5) \Psi \times \bar{\Psi} \gamma^\mu (1 - \gamma^5) \Psi \quad (15)$$

has coupling G_F of dimension $[G_G] = E^{-2}$; its value is $G_F \approx 1.17 \cdot 10^{-5} \text{ GeV}^{-2}$. This is a good effective theory for low-energy weak interactions, but it cannot be used for energies $\mathcal{E} \gtrsim 1/\sqrt{G_F} \sim 300 \text{ GeV}$, not even theoretically. In real life, the Fermi theory works only for $\mathcal{E} \ll M_W \sim 80 \text{ GeV}$; at higher energies, one should use the complete $SU(2) \times U(1)$ electroweak theory instead of the Fermi theory.

Similar to the Fermi theory, most effective theories with $\Delta < 0$ couplings are low-energy limits of more complicated theories with extra heavy particles of masses $M \lesssim g^{-1/\Delta}$ but no $\Delta < 0$ couplings.

In QFTs which are valid for all energies, all coupling must have zero or positive energy dimensions. In 4D, a coupling involving b bosonic fields (scalar or vector), f fermionic fields, and δ derivatives ∂_μ has dimensionality

$$\Delta = 4 - b - \frac{3}{2}f - \delta. \quad (16)$$

Thus, only the boson³ couplings have $\Delta > 0$ while the $\Delta = 0$ couplings comprise boson⁴, boson $\times$ fermion², and boson² $\times$ ∂ boson. All other coupling types have $\Delta < 0$ and are not allowed (except in effective theories).

Here is the complete list of the allowed couplings in 4D.

1. Scalar couplings

$$-\frac{\mu}{3!} \Phi^3 \quad \text{and} \quad -\frac{\lambda}{4!} \Phi^4. \quad (17)$$

Note: the higher powers Φ^5 , Φ^6 , *etc.*, are not allowed because the couplings would have $\Delta < 0$.

2. Gauge couplings of vectors to charged scalars

$$-iqA^\mu (\Phi^* \partial_\mu \Phi - \Phi \partial_\mu \Phi^*) + q^2 \Phi^* \Phi A_\mu A^\mu \subset D_\mu \Phi^* D^\mu \Phi. \quad (18)$$

3. Non-abelian gauge couplings between the vector fields

$$-gf^{abc}(\partial_\mu A_\nu^a)A^{\mu b}A^{\nu c} - \frac{g^2}{4}f^{abc}f^{ade}A_\mu^b A_\nu^c A^{\mu d}A^{\nu e} \subset -\frac{1}{4}F_{\mu\nu}^a F^{\mu\nu a}. \quad (19)$$

4. Gauge couplings of vectors to charged fermions,

$$-qA^\mu \times \bar{\Psi} \gamma_\mu \Psi \subset \bar{\Psi} (i\gamma_\mu D^\mu) \Psi. \quad (20)$$

If the fermions are massless and chiral, we may also have

$$-qA_\mu \times \bar{\Psi} \gamma^\mu \frac{1 \mp \gamma^5}{2} \Psi, \quad (21)$$

or in the Weyl fermion language

$$-qA_\mu \times \chi^\dagger \bar{\sigma}_\mu \chi.$$

5. Yukawa couplings of scalars to fermions,

$$-y\Phi \times \bar{\Psi} \Psi \quad \text{or} \quad -iy\Phi \times \bar{\Psi} \gamma^5 \Psi. \quad (22)$$

If parity is conserved, in the first terms Φ should be a true scalar, and in the second term a pseudo-scalar.

In other spacetime dimensions $d \neq 3+1$, a coupling involving b bosonic fields, f fermionic fields, and δ derivatives has energy dimension

$$\Delta = d - b \times \frac{d-2}{2} - f \times \frac{d-1}{2} - \delta = b + \frac{1}{2}f - \delta - \frac{b+f-2}{2} \times d. \quad (23)$$

Since all interactions involve three or more fields, thus $b+f \geq 3$, the dimensionality of any particular coupling always decreases with d . Consequently, there are more perturbatively-allowed couplings with $\Delta \geq 0$ in lower dimensions $d = 2+1$ or $d = 1+1$ but fewer allowed couplings in higher dimensions $d > 3+1$. In particular,

- In $d \geq 6 + 1$ dimensions all couplings have $\Delta < 0$ and there are no UV-complete quantum field theories, or at least no perturbative UV-complete quantum field theories.
- In $d = 5 + 1$ dimensions there is a unique $\Delta = 0$ coupling $(\mu/6)\Phi^3$, while all the other couplings have $\Delta < 0$. Consequently, the only perturbative UV-complete theories are scalar theories with cubic potentials,

$$\mathcal{L} = \sum_a \left(\frac{1}{2} (\partial_\mu \Phi_a)^2 - \frac{1}{2} m_a^2 \Phi_a^2 \right) - \frac{1}{6} \sum_{a,b,c} \mu_{abc} \Phi_a \Phi_b \Phi_c. \quad (24)$$

However, while such theories are perturbatively OK, they do not have stable vacua. Indeed, a cubic potential is un-bounded from below — it goes to $-\infty$ along half of the directions in the field space — so even if it has a *local* minimum at $\Phi_a = 0$, it's not the global minimum. Consequently, in the quantum theory, the naive vacuum with $\langle \Phi_a \rangle = 0$ would decay by tunneling to a run-away state with $\langle \Phi_a \rangle \rightarrow \pm\infty$.

- In $d = 4 + 1$ dimensions, the $(\mu/6)\Phi^3$ coupling has positive $\Delta = +\frac{1}{2}$ while all the other couplings have negative energy dimensions. Again, the only perturbative UV-complete theories are scalar theories with cubic potentials, but they do not have stable vacua.
- ★ The bottom line is, *in $d > 3 + 1$ dimensions, all quantum field theories are effective theories* for low-enough energies. At higher energies, a different kind of theory must take over — perhaps a theory in a discrete space, perhaps a string theory, or maybe something more exotic.

On the other hand, in lower dimensions $d = 2 + 1$ or $d = 1 + 1$ there are a lot of allowed couplings with $\Delta \geq 0$. In particular, in $d = 2 + 1$ dimensions the allowed couplings include:

- Scalar couplings $(C_n/n!)\Phi^n$ up to $n = 6$;
- gauge and Yukawa couplings like in 4D;
- Yukawa-like couplings $\tilde{y}\Phi^2 \times \bar{\Psi}\Psi$ involving 2 scalars;
- gauge-like couplings with g_{gauge} linearly dependent on a neutral scalar field:

$$D_\mu \Psi = \partial_\mu \Psi + i(g_0 + c\phi)A_\mu \Psi \implies \bar{\Psi}(i\gamma^\mu D_\mu)\Psi \supset -c\phi A_\mu \bar{\Psi}\gamma^\mu \Psi, \quad (25)$$

and likewise

$$D^\mu \Phi^* D_\mu \Phi \supset -ic\phi A^\mu \times (\Phi^* \partial_\mu \Phi - \Phi \partial_\mu \Phi^*) + c^2 \phi^2 \times \Phi^* \Phi A_\mu A^\mu, \quad (26)$$

or non-abelian

$$-\frac{1}{4} F_{\mu\nu}^a F^{\mu\nu a} \supset -c\phi \times f^{abc} (\partial_\mu A_\nu^a) A^{\mu b} A^{\nu c} - \frac{c^2}{4} \phi^2 \times f^{abc} f^{ade} A_\mu^b A_\nu^c A^{\mu d} A^{\nu e}. \quad (27)$$

There are also combinations of $g = g_0 + c\phi$ with Chern–Simons couplings (like the 3D photon mass term in the mid-term exam and its non-abelian generalizations), but I don't want to get into their details here.

* There might be some other allowed couplings in 3D, but never mind for now.

Finally, *in $d = 1 + 1$ dimensions there is an infinite number of allowed $\Delta \geq 0$ couplings.* Indeed, for $d = 1 + 1$ the bosonic fields have energy dimension E^0 , so Δ of a coupling does not depend on the number b of bosonic fields it involves but only on the numbers of derivatives and fermionic fields,

$$\Delta = 2 - \delta - \frac{1}{2}f. \quad (28)$$

Consequently, all scalar potentials $V(\Phi)$ — including $C_n \Phi^n$ terms for any n , and even the non-polynomial potentials — have $\Delta = +2$, so any $V(\Phi)$ is allowed in 2D. Likewise, all Yukawa-like couplings $\Phi^n \bar{\Psi} \Psi$ have $\Delta = +1$, so we may have terms like $y_{IJ}(\Phi) \times \bar{\Psi}^I \Psi^J$ for any functions $y_{IJ}(\Phi)$.

At the $\Delta = 0$ level, we are allowed generic Riemannian metrics $g_{ij}(\Phi)$ of the scalar field space, hence field-dependent kinetic terms

$$\mathcal{L}_{\text{kin}} = \frac{1}{2} g_{ij}(\phi) \times \partial^\mu \phi^i \partial_\mu \phi^j, \quad (29)$$

as well as a whole bunch of fermionic terms with arbitrary scalar-dependent coefficients,

$$\begin{aligned} \mathcal{L}_\Psi \supset \frac{1}{4} g_{IJ}(\Phi) \times \bar{\Psi}^I \gamma^\mu \left(i \vec{\partial}_\mu - i \overleftarrow{\partial}_\mu \right) \Psi^J + \Gamma_{IJK}(\Phi) \times \partial_\mu \Phi^k \times \bar{\Psi}^I \gamma^\mu \Psi^J \\ + \frac{1}{2} R_{IJKL}(\Phi) \times \bar{\Psi}^I \gamma^\mu \Psi^J \times \bar{\Psi}^K \gamma_\mu \Psi^L. \end{aligned} \quad (30)$$

In addition, there are gauge couplings with arbitrary scalar dependent $g_{\text{gauge}}(\Phi)$, chiral couplings to Weyl or Majorana–Weyl fermions, *etc.*, *etc.*